

Age-Decay Sanity Study Report

I. Goal

This report documents a fast sanity-check of anchored age decay using a strict non-train boundary and max CTE as the primary metric.

II. Setup

Train maps: Austin, Budapest, Melbourne, Sakhir, Zandvoort.

Eval maps: Nuerburgring, Sochi, Spa.

Cloud latency: 5.

Lambda grid: 0, 1, 4, 16, 64.

Selection metric: max_cte.

Boundary hit: True.

Selected lambda: $\lambda^* = 0.0$.

III. Commands Run

- `lambda_sweep:` PYTHON-
PATH=packages/f110_gym/src:packages/f110_planning/src:packages/f110_scripts/src
.venv/bin/python scripts/benchmarks/sweep_age_decay_lambda.py
-maps Austin,Budapest,Melbourne,Sakhir,Zandvoort
-trials 2 -workers 5 -selection-metric max_cte
-lambda-values 0,1,4,16,64 -sigma-proc-left 0.044961 -sigma-proc-track 0.067937 -sigma-proc-heading 0.033182 -output-stem lambda_sweep_sanity_5train -selected-lambda-json data/benchmarks/lambda_sweep_sanity_optimal.json
- `train_family_search:` PYTHON-
PATH=packages/f110_gym/src:packages/f110_planning/src:packages/f110_scripts/src
.venv/bin/python scripts/benchmarks/optimize_hyperparameters.py
-maps Austin,Budapest,Melbourne,Sakhir,Zandvoort
-trials 1 -workers 5 -selection-metric max_cte -grid-profile sanity -age-decay-lambda 0 -output-stem strategy_sanity_5train
- `heldout_static_full_suite:` PYTHON-
PATH=packages/f110_gym/src:packages/f110_planning/src:packages/f110_scripts/src
.venv/bin/python scripts/benchmarks/benchmark_single_tier_sanity_3eval_static.py
-maps Nuerburgring,Sochi,Spa -trials 3 -workers 3 -selection-metric max_cte -output-stem single_tier_sanity_3eval_static
- `heldout_lambda_full_suite:` Reused single_tier_sanity_3eval_static outputs under single_tier_sanity_3eval_lambda because selected lambda was 0.0, making anchored decay identical to static fusion in this sanity study.
- `heldout_best_configs:` PYTHON-
PATH=packages/f110_gym/src:packages/f110_planning/src:packages/f110_scripts/src
.venv/bin/python scripts/benchmarks/benchmark_eval_best_configs.py
-configs-json data/benchmarks/strategy_sanity_5train_best_configs.json
-maps Nuerburgring,Sochi,Spa -trials 5 -workers 3

```
-selection-metric max_cte -age-decay-lambda 0 -  
output-stem eval_best_configs_sanity_3eval_lambda  
• figure_generation: PYTHON-  
PATH=packages/f110_gym/src:packages/f110_planning/src:packages/f110_scripts/src  
.venv/bin/python scripts/benchmarks/plot_age_decay_sanity_figures.py  
-lambda-raw-csv data/benchmarks/lambda_sweep_sanity_5train_raw.csv  
-selected-lambda-json data/benchmarks/lambda_sweep_sanity_optimal.json  
-static-raw-csv data/benchmarks/single_tier_sanity_3eval_static_raw.csv  
-lambda-raw-heldout-csv data/benchmarks/single_tier_sanity_3eval_lambda_raw.csv  
-best-config-raw-csv data/benchmarks/eval_best_configs_sanity_3eval_raw.csv  
-output-dir data/benchmarks/paper_figures_sanity_3eval
```

IV. Best Train-Time Family Configs

- `always:` always_hit, train max CTE 0.3223, CCR 1.000.
- `bernoulli_max_miss:` bernoulli_max_miss_p65_m6, train max CTE 0.3198, CCR 0.644.
- `deterministic:` deterministic_t0p11, train max CTE 0.3201, CCR 0.867.
- `exponential:` exponential_c0p08_r5, train max CTE 0.3158, CCR 0.655.
- `fixed_bernoulli:` fixed_bernoulli_p65, train max CTE 0.3186, CCR 0.644.
- `fixed_interval:` fixed_interval_k2, train max CTE 0.3237, CCR 0.500.
- `logistic:` logistic_c0p08_s10, train max CTE 0.3224, CCR 0.851.
- `piecewise_ramp:` piecewise_ramp_0p05_0p12, train max CTE 0.3185, CCR 0.899.

V. Held-Out Best-Family Results

Strategy	Experiment	CCR
always	always_hit	1.000
bernoulli_max_miss	bernoulli_max_miss_p65_m6	0.651
deterministic	deterministic_t0p11	0.869
exponential	exponential_c0p08_r5	0.676
fixed_bernoulli	fixed_bernoulli_p65	0.649
fixed_interval	fixed_interval_k2	0.500
logistic	logistic_c0p08_s10	0.864
piecewise_ramp	piecewise_ramp_0p05_0p12	0.904

VI. Figures

Appendix A

Lambda Sweep

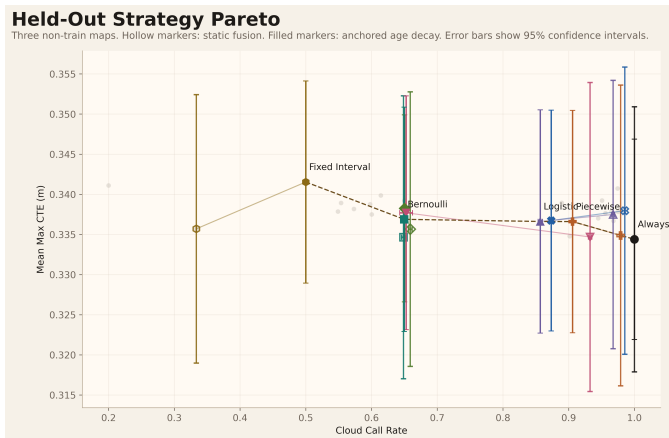


Fig. 1. Aggregate held-out Pareto view.

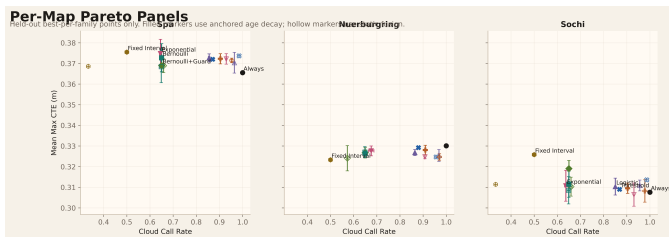


Fig. 2. Per-map Pareto panels for the three eval maps.

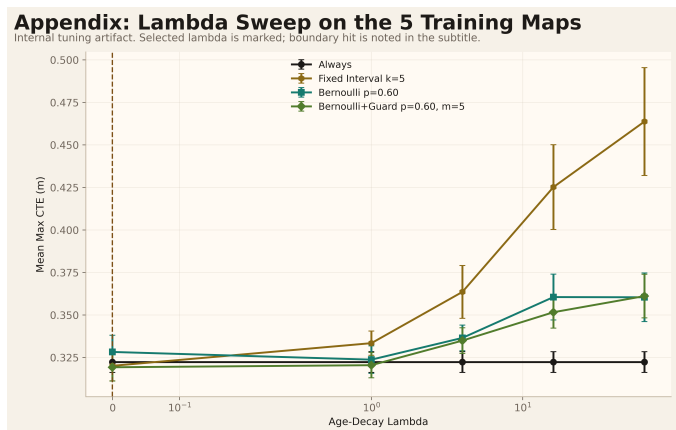


Fig. 4. Internal lambda-sweep appendix figure.

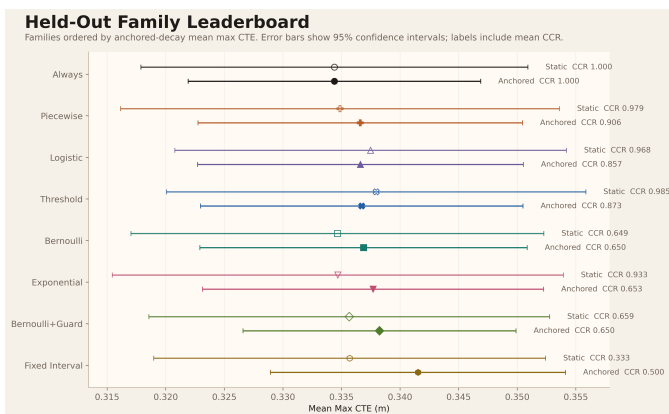


Fig. 3. Family leaderboard on the held-out eval maps.